3. Arithmetic Progression

Q21. The 5th term of an A.P. is 17 and the 12th term is 38. Find the first term and common difference.

Given: $a_5 = 17$, $a_{12} = 38$

Formula: $a_n = a + (n - 1)d$

$a_5 = a + 4d = 17$ ...(1)

$a_{12} = a + 11d = 38$ ...(2)

Subtract (1) from (2): $7d = 21 \Rightarrow d = 3$

Put $d = 3$ in (1): $a + 4(3) = 17 \Rightarrow a + 12 = 17 \Rightarrow a = 5$

Answer: First term $a = 5$, common difference $d = 3$.

Q22. How many terms of the A.P. 7, 10, 13, … are needed to give a sum of 301?

Given A.P.: $a = 7$, $d = 3$, Sum $S_n = 301$

Formula: $S_n = n/2\ [2a + (n - 1)d]$

$301 = n/2\ [2(7) + (n - 1)3]$

$301 = n/2\ [14 + 3n - 3] = n/2\ (3n + 11)$

$\Rightarrow n(3n + 11) = 602$

$\Rightarrow 3n^2 + 11n - 602 = 0$

$n = [-11 + \sqrt{(11^2 + 4 \cdot 3 \cdot 602)}] / (2 \cdot 3)$

$n = [-11 + \sqrt{(121 + 7224)}] / 6 = [-11 + \sqrt{7345}] / 6$

$\approx [-11 + 85.70] / 6 \approx 12.45$

Answer: $n = (-11 + \sqrt{7345})/6 \approx 12.45$ (Hence, 301 is not obtained for an integer number of terms; question data seems inconsistent. The method and result are correct.)

Q23. The nth term of an A.P. is $a_n = 5 - 3n$. Find its 10th term and 25th term.

Given: $a_n = 5 - 3n$

$a_{10} = 5 - 3(10) = 5 - 30 = -25$

$a_{25} = 5 - 3(25) = 5 - 75 = -70$

Answer: 10th term = −25, 25th term = −70.

Q24. Find the sum of first 30 terms of the A.P. 4, 9, 14, …

Given: $a = 4$, $d = 5$, $n = 30$

$S_n = n/2\ [2a + (n - 1)d]$

$S_{30} = 30/2\ [2(4) + 29(5)]$

$= 15\ [8 + 145] = 15 \times 153 = 2295$

Answer: Sum of first 30 terms = 2295.

Q25. In an A.P., the 3rd term is 8 and the 7th term is 20. Find the 12th term and the sum of first 12 terms.

Given: $a_3 = 8$, $a_7 = 20$

$a_3 = a + 2d = 8$ ...(1)

$a_7 = a + 6d = 20$ ...(2)

Subtract (1) from (2): $4d = 12 \Rightarrow d = 3$

Put $d = 3$ in (1): $a + 2(3) = 8 \Rightarrow a + 6 = 8 \Rightarrow a = 2$

12th term: $a_{12} = a + 11d = 2 + 11(3) = 35$

Sum of first 12 terms: $S_{12} = 12/2\ [2a + 11d]$

$= 6\ [2(2) + 11(3)] = 6\ [4 + 33] = 6 \times 37 = 222$

Answer: $a_{12} = 35$, $S_{12} = 222$.

Q26. The sum of first n terms of an A.P. is $S_n = 2n^2 + 3n$. Find (i) the first term (ii) the 10th term.

Given: $S_n = 2n^2 + 3n$

First term: $a_1 = S_1 = 2(1)^2 + 3(1) = 2 + 3 = 5$

nth term: $a_n = S_n - S_{n-1}$

$S_{n-1} = 2(n - 1)^2 + 3(n - 1) = 2(n^2 - 2n + 1) + 3n - 3$

$= 2n^2 - 4n + 2 + 3n - 3 = 2n^2 - n - 1$

Thus $a_n = (2n^2 + 3n) - (2n^2 - n - 1) = 4n + 1$

So, $a_{10} = 4(10) + 1 = 41$

Answer: (i) First term $a = 5$ (ii) 10th term = 41.

Q27. A man saves ₹200 in the first month, ₹260 in the second month, ₹320 in the third month and so on. In how many months will he save ₹9,720 in total?

Given savings form an A.P.: $a = 200$, $d = 60$, total $S_n = 9720$

$S_n = n/2\ [2a + (n - 1)d]$

$9720 = n/2\ [2(200) + (n - 1)60]$

$9720 = n/2\ [400 + 60n - 60] = n/2\ (60n + 340)$

$\Rightarrow 9720 = n(30n + 170)$

$\Rightarrow 30n^2 + 170n - 9720 = 0$

Divide by 10: $3n^2 + 17n - 972 = 0$

$n = [-17 + \sqrt{(17^2 + 4 \cdot 3 \cdot 972)}] / (2 \cdot 3)$

$n = [-17 + \sqrt{(289 + 11664)}] / 6 = [-17 + \sqrt{11953}] / 6$

$\approx [-17 + 109.33] / 6 \approx 15.39$

Answer: $n = (-17 + \sqrt{11953})/6 \approx 15.39$ (Hence, 9720 is not obtained for an integer number of months; the given total appears inconsistent. The method and result are correct.)

Q28. The 9th term of an A.P. is zero. If its 5th term is 12, find its first term and common difference.

Given: $a_9 = 0$, $a_5 = 12$

$a_9 = a + 8d = 0$ ...(1)

$a_5 = a + 4d = 12$ ...(2)

Subtract (2) from (1): $4d = -12 \Rightarrow d = -3$

Put $d = -3$ in (2): $a + 4(-3) = 12 \Rightarrow a - 12 = 12 \Rightarrow a = 24$

Answer: First term $a = 24$, common difference $d = -3$.

Q29. The sum of first 15 terms of an A.P. is 525 and the sum of the next 15 terms is 1200. Find the first term and common difference.

Given: $S_{15} = 525$ and $(S_{30} - S_{15}) = 1200 \Rightarrow S_{30} = 525 + 1200 = 1725$

Formula: $S_n = n/2\,[2a + (n - 1)d]$

$S_{15} = 15/2\,[2a + 14d] = 525 \Rightarrow 15[2a + 14d] = 1050 \Rightarrow 2a + 14d = 70$ ...(1)

$S_{30} = 30/2\,[2a + 29d] = 15[2a + 29d] = 1725 \Rightarrow 2a + 29d = 115$ ...(2)

Subtract (1) from (2): $15d = 45 \Rightarrow d = 3$

Put $d = 3$ in (1): $2a + 14(3) = 70 \Rightarrow 2a + 42 = 70 \Rightarrow 2a = 28 \Rightarrow a = 14$

Answer: First term $a = 14$, common difference $d = 3$.

Q30. The interior angles of a polygon are in A.P. with common difference 5°. The smallest angle is 120° and the largest is 170°. How many sides has the polygon?

Given: smallest angle $a = 120°$, common difference $d = 5°$, largest angle $= 170°$

Let number of angles (sides) = n. Then last term: $a_n = a + (n - 1)d$

$170 = 120 + (n - 1)5$

$50 = 5(n - 1) \Rightarrow n - 1 = 10 \Rightarrow n = 11$

Answer: The polygon has 11 sides.